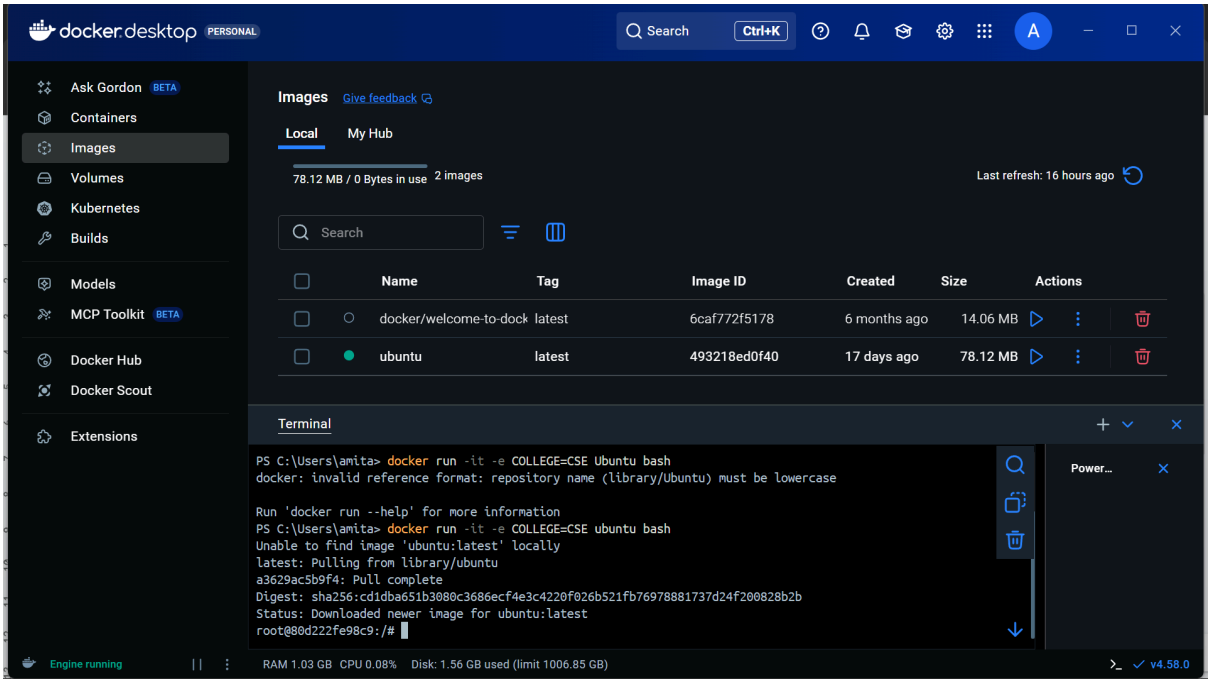
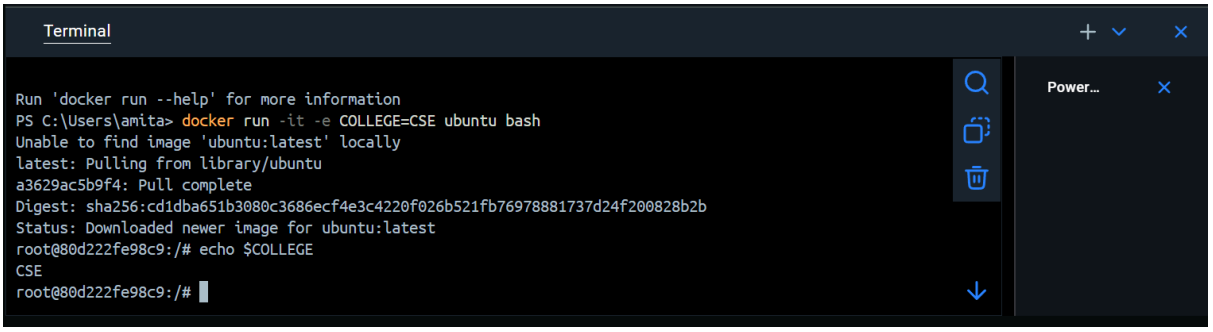
Task #1
Run Ubuntu container
Pass -e COLLEGE=CSE
Show echo $COLLEGE
Stop container then check what happened



Commands: docker run -it -e COLLEGE=CSE ubuntu bash



Commands: echo $COLLEGE

```
Terminal
PS C:\Users\amita> docker run -it -e COLLEGE=CSE ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
a3629ac5b9f4: Pull complete
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Downloaded newer image for ubuntu:latest
root@80d222fe98c9:/# echo $COLLEGE
CSE
root@80d222fe98c9:/# exit
exit
PS C:\Users\amita>
RAM 1.04 GB  CPU 0.00%  Disk: 1.56 GB used (limit 1006.85 GB)
Power...
v4.58.0
```

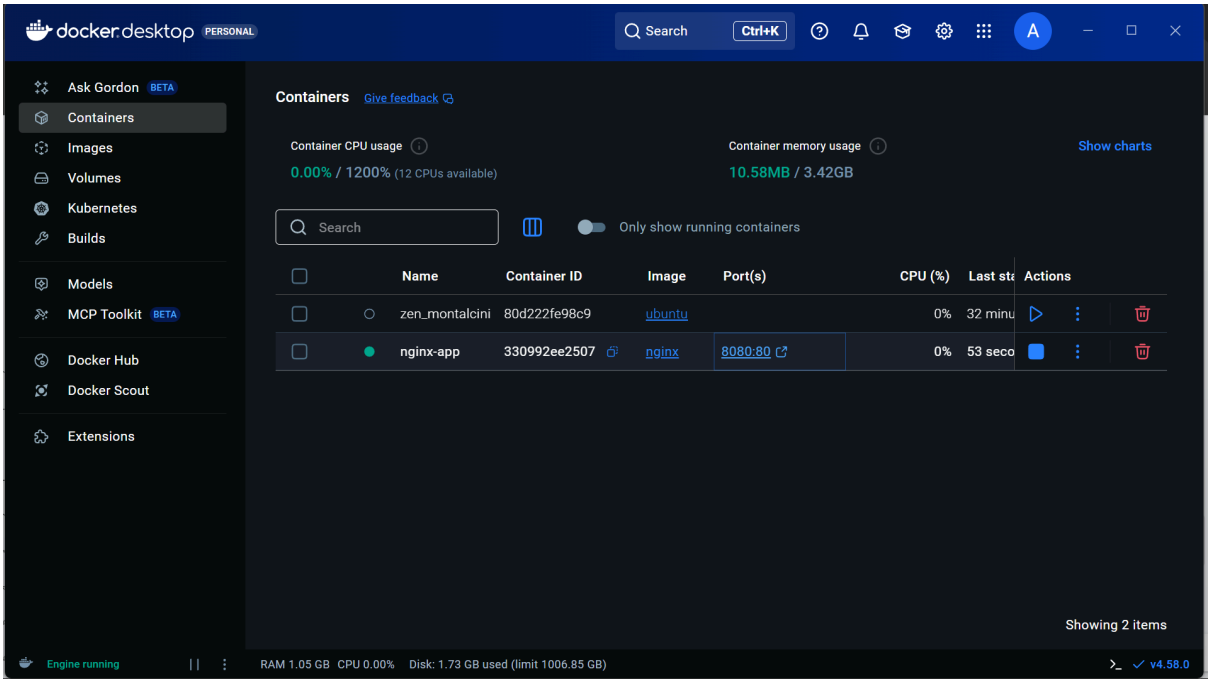
Commands : exit or docker stop root@80d222fe98c9

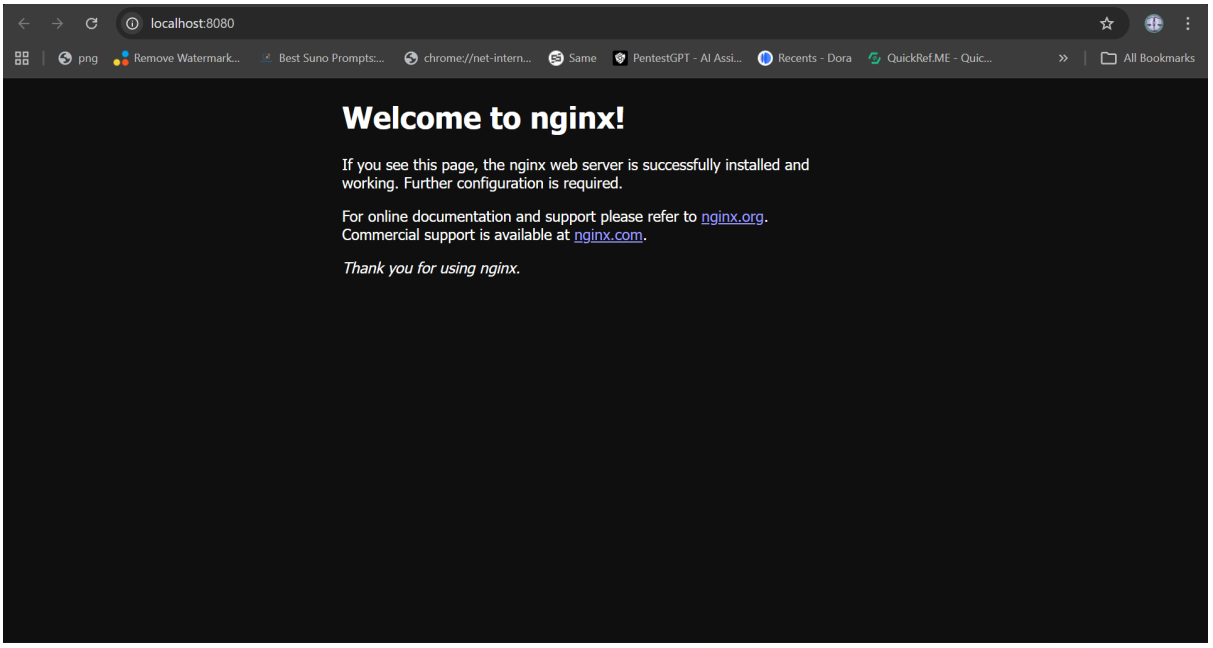
Practice Questions on Basic Docker Commands

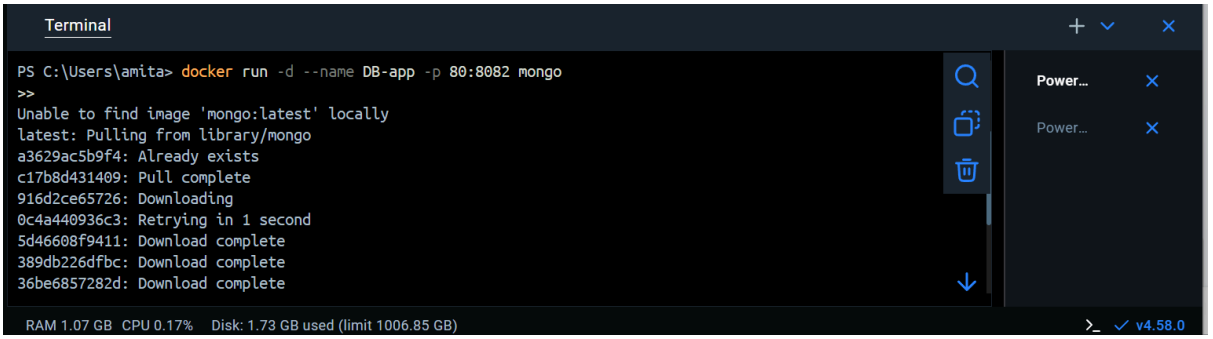
Run a Docker container named “DB-app” based on the “mongodb” image, and expose port 80 on the host to port 8082 on the container?

```
PS C:\Users\amita> docker run -it --name nginx-app -p 8080:80 -e NGINX_PORT=8080 nginx
>>
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
RAM 1.05 GB  CPU 0.25%  Disk: 1.73 GB used (limit 1006.85 GB)
Power...
v4.58.0
```

Run a Docker container based on the nginx image, exposing port 8080 on the host port 80 on the container . Set an environment variable NGINX_PORT=8080 inside the container and start the container interactively.


docker desktop PERSONAL
Search Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers Give feedback
Container CPU usage
0.00% / 1200% (12 CPUs available)
Container memory usage
10.58MB / 3.42GB
Show charts
Search
Only show running containers
Name Container ID Image Port(s) CPU (%) Last sta Actions
zen_montalcini 80d222fe98c9 ubuntu 0% 32 minu
nginx-app 330992ee2507 nginx 8080:80 0% 53 seco
Showing 2 items
Engine running
RAM 1.05 GB CPU 0.00% Disk: 1.73 GB used (limit 1006.85 GB)
v4.58.0


localhost:8080
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.


Terminal
PS C:\Users\amita> docker run -d --name DB-app -p 80:8082 mongo
>>
Unable to find image 'mongo:latest' locally
latest: Pulling from library/mongo
a3629ac5b9f4: Already exists
c17b8d431409: Pull complete
916d2ce65726: Downloading
0c4a440936c3: Retrying in 1 second
5d46608f9411: Download complete
389db226dfbc: Download complete
36be6857282d: Download complete
Power...
Power...
RAM 1.07 GB CPU 0.17% Disk: 1.73 GB used (limit 1006.85 GB)
v4.58.0

```
Terminal
5d46608f9411: Download complete
389db226dfbc: Download complete
36be6857282d: Download complete
2f646e1be2da: Download complete
docker: Get "https://registry-1.docker.io/v2/library/mongo/blobs/sha256:0c4a440936c320dc21725487831f112a5
3d43704f67ae4c7059b14d88be53655": dialing registry-1.docker.io:443 container via direct connection becaus
e Docker Desktop has no HTTPS proxy: connecting to registry-1.docker.io:443: dial tcp: lookup registry-1.
docker.io: no such host

Run 'docker run --help' for more information
PS C:\Users\amita>
RAM 1.06 GB  CPU 0.00%  Disk: 1.73 GB used (limit 1006.85 GB)
Power...
Power...
v4.58.0
```

```
PS C:\Users\amita> docker run -it --name nginx-app -p 8080:80 -e NGINX_PORT=8080 nginx
>>
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
RAM 1.06 GB  CPU 0.00%  Disk: 1.73 GB used (limit 1006.85 GB)
Power...
v4.58.0
```

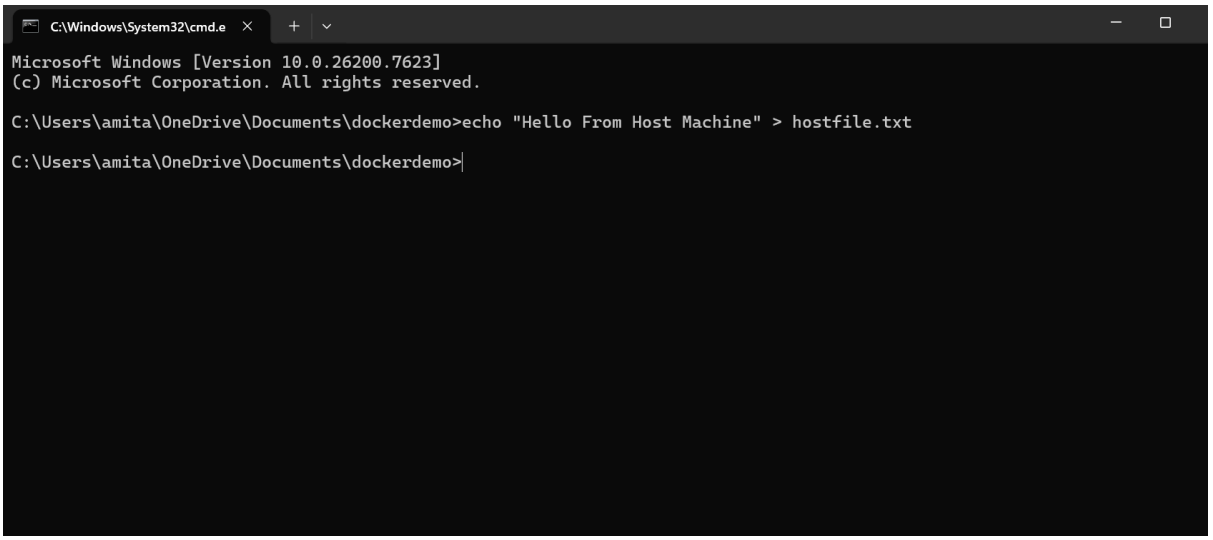
3. How would you use the docker run command with -it , -e , -v and –name to:
   Set an environment variable APP_ENV=production.
  Name the container my_app.
 Start an interactive terminal in an image called my_image?

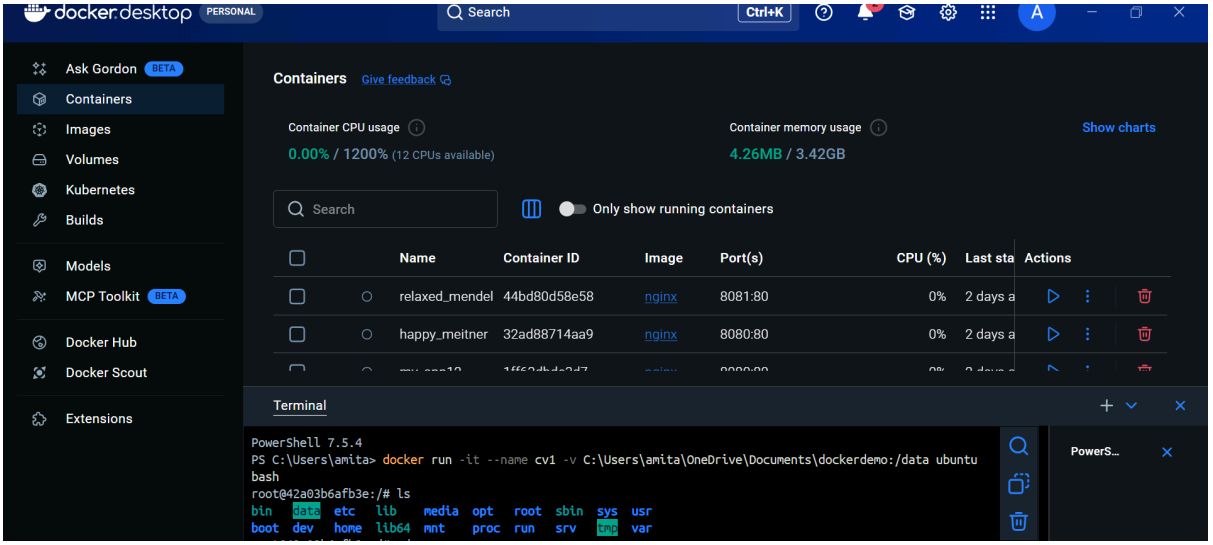
Write the full commands for the scenario above.

Commands: docker run -it --name my_app -e APP_ENV=production -v $(pwd):/app my_image /bin/bash

Example 1
Mkdir C:\docker-volume-demo
Echo “hello from HOST machine > C:\docker-volume-demo\hostfile.txt
Docker run -it --name cv1 -v c:\docker-volume-demo:/data ubuntu bash

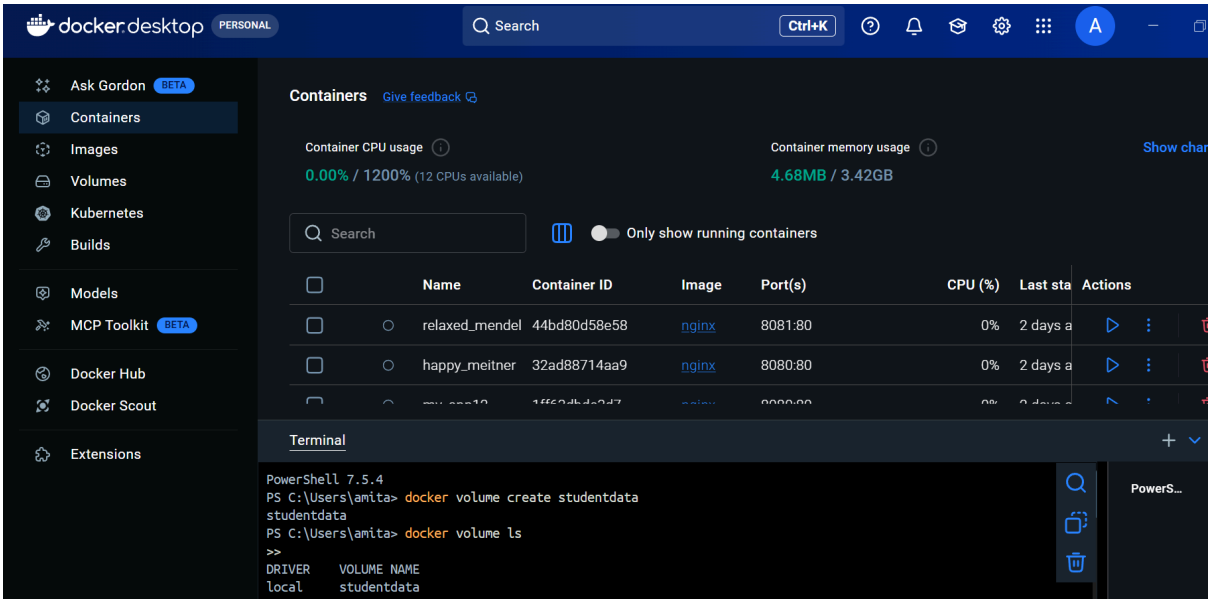
Container
ls
Cd data
Cat hostfile.txt
Modify file inside container
Echo “Modified inside container” >> /data/hostfile.txt
Exit





Practice Question

Create a Docker volume named studentdata.
Run an Ubuntu container and mount studentdata at /student.
Create a file inside the container and verify it persists after container deletion.
Attach the same volume to another container and verify the file exists.
Demonstrate data sharing between two containers using a shared volume.
Show that deleting a container does NOT delete the volume data.



docker run -d -p 80:80 --name my_container nginx
Start a stopped container
docker start<container_id/name>
unpause a paused container
docker unpause<container_id/name>
Remove a container
docker rm<container_id/name>
Remove all stopped container

List all running container
docker ps
List all containers
docker ps -a
Show detailed info about a container
docker inspect<container_id/name>
Display resource uage stats of container
docker task<conatiner_id/name>

Execute a commands
docker exec -it <container_id/name><command>
Attach to a running container

docker attach<container_id/name>
Copy file
from container to host:
docker cp <container_id>container_path>

-p optional flag

Docker Volumes
A Docker volume is a persistent storage mechanism used by containers to store data outside the container's writable layer.
Key idea:
Containers are temporary, volumes are permanent

work volumes
docker volume create <volume name>
docker volume ls
docker volume inspect mydata
docker run -d -v

store data in a volume

docker run -it -v <source>:<path of destination><image name>sh
cat<path of file>

To remove a volume

Docker volume rm<volume name>
to delete all unsent volumes
docker volume prune

Example 1
1. mkdir C:\docker-volume-demo
2. echo HELLO from HOST machine > C:\docker-volume-demo\hostfile.txt
3. docker run -it --name cv1 -v c:\docker-volume-demo:/data ubuntu bash
4. Inside container
ls
cd data
cat hostfile.txt
5. Modify file Inside container
echo "Modified inside container" >>/data/hostfile.txt
exit

tag the Docker Image
docker tag<image-id> <docker-hub-username>/<image-name>:<tag>

1.Topic:
To push the update image and to update then push image on docker hub.
1. pull any base image on host machine
2. To tag that base image with docker hub username
3. docker login
4. docker push updated image

2. topic
Update the image by making changes through container than push to Docker hub.
two part 1. from base image 2. from updated image
we need to do
1. run the container through B.I/U.I and then interact the container.
2. Inside container ---- do changes.
3. Then commit the changes in in container through image.
4. Push that image an docker hub.
5. Need to verify the changes by running another container from same image.
6. Also verify by compare the size of image with previous image.

docker commit <imagename> <username>/<imageid>
steps :
docker pull alpine
docker tag alpine username/imagename:tag
docker login
docker run -it --name a1 username/imagename:tag
apt update
php -v
exit
5. --- docker stop a1
6. --- docker rm a1
7. -- docker run -it --name a2 username/imagename:v1.1
 php -v
exit
8.---- docker push username/imagename:v1.1

Step-4 Verify and connect to container
 Docker ps
Docker exec -it mysql1 bash

Step-5

Mysql -h 127.0.0.1 -u myuser -p
Or directly
Docker exec -it musql1 mysql -u root -p

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\amita> pwd

Path

C:\Users\amita

```
PS C:\Users\amita> ls


    Directory: C:\Users\amita


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13-01-2026     15:15                .antigravity
d-----        29-01-2026     20:34                .aws
d-----        29-01-2026     20:34                .azure
d-----        30-01-2026     22:46                .cache
d-----        23-01-2026     12:40                .cagent
d-----        23-01-2026     12:40                .config
d-----        13-02-2026     12:09                .docker
d-----        13-01-2026     20:22                .gemini
d-----        24-11-2025     21:33                .ipython
d-----        08-02-2026     01:56                .keras
d-----        05-12-2025     12:56                .m2
d-----        08-02-2026     01:27                .matplotlib
d-----        21-01-2026     20:53                .ollama
d-----        27-12-2025     14:54                .streamlit
d-----        21-11-2025     12:03                .VirtualBox
d-----        16-11-2025     19:28                .vscode
d-r---        13-11-2025     23:59                Contacts
d-----        03-12-2025     11:57                Documents
dar---        11-02-2026     23:27                Downloads
d-r---        13-11-2025     23:59                Favorites
d-----        10-12-2025     14:28                fileak
d-r---        13-11-2025     23:59                Links
d-r---        13-11-2025     23:59                Music
d-----        11-12-2025     12:36                my-app
d-----        11-12-2025     10:28                myproject
dar--l        09-02-2026     20:04                OneDrive
d-----        10-02-2026     00:02                Pictures
d-r---        13-11-2025     23:59                Saved Games
d-r---        14-11-2025     00:01                Searches
d-----        10-12-2025     12:34                student
d-----        17-12-2025     15:46                TicketBoard
d-r---        08-02-2026     18:20                Videos
d-----        16-11-2025     13:28                VirtualBox VMs
-a----        26-12-2025     12:54           2430 .bash_history
-a----        14-01-2026     00:17            236 .gitconfig
-a----        27-12-2025     12:21             20 .lesshst
-a----        10-12-2025     14:16             66 .minttyrc
-a----        20-01-2026     23:52          80643 index.html
-a----        21-01-2026     00:01             13 README.md
```

```
PS C:\Users\amita> cd D:\docker
PS D:\docker> mkdir mysql


    Directory: D:\docker


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13-02-2026     12:29                mysql


PS D:\docker> nano docker file
nano : The term 'nano' is not recognized as the name of a cmdlet, function, script file, or
operable
program. Check the spelling of the name, or if a path was included, verify that the path is
correct and
try again.
At line:1 char:1
+ nano docker file
+ ~~~~
    + CategoryInfo          : ObjectNotFound: (nano:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS D:\docker> nano dockerfile
nano : The term 'nano' is not recognized as the name of a cmdlet, function, script file, or
operable
program. Check the spelling of the name, or if a path was included, verify that the path is
correct and
try again.
At line:1 char:1
+ nano dockerfile
+ ~~~~
    + CategoryInfo          : ObjectNotFound: (nano:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS D:\docker> notepad dockerfile
PS D:\docker> docker build -t mysql1
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
PS D:\docker> docker build -t mysql .
[+] Building 0.3s (1/1) FINISHED                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                        0.1s
```

```
 => => transferring dockerfile: 2B                                                    0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file
or directory
PS D:\docker> docker build -t mysql .
[+] Building 0.1s (1/1) FINISHED                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                   0.0s
 => => transferring dockerfile: 2B                                                    0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file
or directory
PS D:\docker> cd mysql
PS D:\docker\mysql> docker build -t mysql1 .
[+] Building 181.3s (6/6) FINISHED                                        docker:desktop-linux
 => [internal] load build definition from dockerfile                                   0.1s
 => => transferring dockerfile: 187B                                                  0.0s
 => WARN: ConsistentInstructionCasing: Command 'From' should match the case of the
command majority  0.1s
 => [internal] load metadata for docker.io/library/mysql:8                             9.4s
 => [auth] library/mysql:pull token for registry-1.docker.io                           0.0s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/1] FROM
docker.io/library/mysql:8@sha256:7fcf7bcd3fa703ff23a2b998e4ed9077e5db570bcc4468459
171.5s
 => => resolve
docker.io/library/mysql:8@sha256:7fcf7bcd3fa703ff23a2b998e4ed9077e5db570bcc4468459
90  0.0s
 => => sha256:7fcf7bcd3fa703ff23a2b998e4ed9077e5db570bcc4468459902e33b23d2841d
2.60kB / 2.60kB       0.0s
 => => sha256:683eb124e222ffca894ba53f35c84bfd43e3e6343664ca909c09a8a45aeffc23
2.86kB / 2.86kB       0.0s
 => =>
sha256:c562866f17cc749b3082dc8a1602ea707a134145679f048a3c0f41c9e36713a0
6.68kB / 6.68kB       0.0s
 => => sha256:d3202636d61f308e37499c87644535b315fa6d989c7fdbf9140ef480ff570d07
885B / 885B           1.7s
 => =>
sha256:4f37333d1be658a226cdafd622c7bda0a95abbcb999c29a571136add51950044
47.31MB / 47.31MB    120.7s
 => => sha256:6ccfd188136ea5c39fd741aaa68c04a1312bca49803a517c8a1ff4e9ae94832a
783.56kB / 783.56kB   4.0s
 => =>
sha256:1e9455481d2b938256b11b624ecd8eab2fbf002c132d14c95fa19b6315066d5b
6.17MB / 6.17MB       15.2s
 => =>
sha256:75882d54c3d8839f01970b3adb9e3d007ff23bd64437aefaceea6e2294ead050
2.61kB / 2.61kB       5.2s
```

```
 => =>
sha256:7a2e50d8b4d194b488b0a9645af73c0fb4927d433780a589b7d06c86e9ef9a07 334B
/ 334B           6.3s
 => => sha256:edb84af18ccbf7a3575f64e23770f09e23be35fec89d896e42e4f5c3154ef765
47.81MB / 47.81MB    79.0s
 => => sha256:ed7fb090b90f53c38ba9bb2b1b0f3f56f9e2b620fc4bf72d55b4f906481d8d53
321B / 321B          16.2s
 => => sha256:9c2c353aa8db811ce8bcbd9414dc87c177fa42b8d326e951f981bd47ec0c1e4
131.14MB / 131.14MB  148.8s
 => =>
sha256:952b63b654ecc06674b303b2e2886751b6cd7266e41b6306bb769109bd9e8e44
5.33kB / 5.33kB      80.9s
 => => extracting
sha256:4f37333d1be658a226cdafd622c7bda0a95abbcb999c29a571136add51950044
5.1s
 => => extracting
sha256:d3202636d61f308e37499c87644535b315fa6d989c7fdbf9140ef480ff570d07
0.0s
 => => extracting
sha256:6ccfd188136ea5c39fd741aaa68c04a1312bca49803a517c8a1ff4e9ae94832a
0.0s
 => => extracting
sha256:1e9455481d2b938256b11b624ecd8eab2fbf002c132d14c95fa19b6315066d5b
0.7s
 => => extracting
sha256:75882d54c3d8839f01970b3adb9e3d007ff23bd64437aefaceea6e2294ead050
0.0s
 => => extracting
sha256:7a2e50d8b4d194b488b0a9645af73c0fb4927d433780a589b7d06c86e9ef9a07
0.0s
 => => extracting
sha256:edb84af18ccbf7a3575f64e23770f09e23be35fec89d896e42e4f5c3154ef765
2.3s
 => => extracting
sha256:ed7fb090b90f53c38ba9bb2b1b0f3f56f9e2b620fc4bf72d55b4f906481d8d53
0.0s
 => => extracting
sha256:9c2c353aa8db811ce8bcbd9414dc87c177fa42b8d326e951f981bd47ec0c1e48
22.2s
 => => extracting
sha256:952b63b654ecc06674b303b2e2886751b6cd7266e41b6306bb769109bd9e8e44
0.0s
 => exporting to image                                                    0.1s
 => => exporting layers                                                   0.0s
 => => writing image
sha256:356678ab0c8df6d03485054782a06dd71dc26b0ff56815473ca3eb045e9f53c8
0.0s
 => => naming to docker.io/library/mysql1                                 0.0s
```

```
 3 warnings found (use docker --debug to expand):
 - ConsistentInstructionCasing: Command 'From' should match the case of the command
majority (uppercase) (line 1)
 - SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV
"MYSQL_ROOT_PASSWORD") (line 2)
 - SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV
"MYSQL_PASSWORD") (line 5)
PS D:\docker\mysql> docker build -t mysql1 .
[+] Building 1.8s (5/5) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from dockerfile                                0.0s
 => => transferring dockerfile: 187B                                            0.0s
 => [internal] load metadata for docker.io/library/mysql:8                              1.7s
 => [internal] load .dockerignore                                               0.0s
 => => transferring context: 2B                                                 0.0s
 => CACHED [1/1] FROM
docker.io/library/mysql:8@sha256:7fcf7bcd3fa703ff23a2b998e4ed9077e5db570bcc44  0.0s
 => exporting to image                                                          0.0s
 => => exporting layers                                                         0.0s
 => => writing image
sha256:356678ab0c8df6d03485054782a06dd71dc26b0ff56815473ca3eb045e9f53c8
0.0s
 => => naming to docker.io/library/mysql1                                           0.0s

 2 warnings found (use docker --debug to expand):
 - SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV
"MYSQL_ROOT_PASSWORD") (line 2)
 - SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV
"MYSQL_PASSWORD") (line 5)
PS D:\docker\mysql> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS D:\docker\mysql> docker exec -it mysql1 bash
Error response from daemon: No such container: mysql1
PS D:\docker\mysql> ^C
PS D:\docker\mysql> docker run -d \
docker: invalid reference format

Run 'docker run --help' for more information
PS D:\docker\mysql> --name mysql1 \
At line:1 char:3
+ --name mysql1 \
+   ~
Missing expression after unary operator '--'.
At line:1 char:3
+ --name mysql1 \
+   ~~~~
Unexpected token 'name' in expression or statement.
    + CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
```

```
    + FullyQualifiedErrorId : MissingExpressionAfterOperator

PS D:\docker\mysql> -p 3306:3306 \
-p : The term '-p' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is
correct and
try again.
At line:1 char:1
+ -p 3306:3306 \
+ ~~
    + CategoryInfo          : ObjectNotFound: (-p:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS D:\docker\mysql> mysql1
mysql1 : The term 'mysql1' is not recognized as the name of a cmdlet, function, script file, or
operable
program. Check the spelling of the name, or if a path was included, verify that the path is
correct and
try again.
At line:1 char:1
+ mysql1
+ ~~~~~~
    + CategoryInfo          : ObjectNotFound: (mysql1:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS D:\docker\mysql> docker run -d \ --name mysql1 \-p 3306:3306 \ mysql1
docker: invalid reference format

Run 'docker run --help' for more information
PS D:\docker\mysql> docker run -d -p 3306:3306 --name mysql2 mysql1
90f169ea78e000a53d807a3f93cda356ffcbba56275f500343520642ffe282db
PS D:\docker\mysql> docker exec -it mysql2 bash
bash-5.1# mysql -h 127.0.0.1 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.8 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Pratical No 3
WorkDir
Sets the default directory inside container
If directory does not exist —-------- docker creates it
All next commands execute inside this folder
COPY commands
Copies files/folders from host container
Simple file transfer
No extra processing
ADD command
Copies files
Auto extracts archives
Downloads files from URLs

RUN command
Excutes commands during image build
Create new Docker layer
Excute commands while building image

Next is:
1. Create Python application
   Docker_python inside;
      app.py , requirement.txt, dockerfile
2. Docker build -t python11 .
3. Docker image
4. Docker run -d –name p1 -p5001:host port python11
 In side APP.py
From flask import flask

Inside requirement.txt

Inside Dockerfile

1. Base image
2. Set working directory
3. Copy dependency file
4. Install dependenciesR
5. Copy Application code
6. Expose port
7. Run the app

docker desktop PERSONAL

Search Ctrl+K

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Builds
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MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions

Containers Give feedback

Container CPU usage
1.91% / 1200% (12 CPUs available)

Container memory usage
131.77MB / 3.42GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last star	Actions
mystifying_feist	094f01b6e3ef	python11	5001:5001	0.02%	2 minute	
interveiw	-	-	-	1.88%	22 minut	

Terminal

```
PS D:\docker\dockerpython> notepad server.py
PS D:\docker\dockerpython> notepad requirements.txt
PS D:\docker\dockerpython> notepad dockerfile
PS D:\docker\dockerpython> docker build -t python11 .
[+] Building 0.2s (1/1) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                0.1s
 => => transferring dockerfile: 2B                                                  0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or direc
tory
PS D:\docker\dockerpython> dir
```

Windo...

Engine running | RAM 1.03 GB CPU 0.08% Disk: 8.88 GB used (limit 1006.85 GB) | v4.60.1

docker desktop PERSONAL

Search Ctrl+K

Ask Gordon
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Extensions

Containers Give feedback

Container CPU usage
1.12% / 1200% (12 CPUs available)

Container memory usage
131.87MB / 3.42GB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last star	Actions
mystifying_feist	094f01b6e3ef	python11	5001:5001	0.03%	2 minute	
interveiw	-	-	-	1.76%	23 minut	

Terminal

```
PS D:\docker\dockerpython> ren dockerfile.txt Dockerfile
PS D:\docker\dockerpython> dir


    Directory: D:\docker\dockerpython


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        18-02-2026     12:25            326 Dockerfile
```

Windo...

Engine running | RAM 1.07 GB CPU 4.26% Disk: 8.88 GB used (limit 1006.85 GB) | v4.60.1

Terminal

```
PS D:\docker\dockerpython> docker build -t python11 .
[+] Building 19.3s (11/11) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 365B                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                   3.1s
 => [auth] library/python:pull token for registry-1.docker.io                         0.0s
 => [internal] load .dockerignore                                                     0.1s
 => => transferring context: 2B                                                       0.0s
 => CACHED [1/5] FROM docker.io/library/python:3.10-slim@sha256:e508a34e5491225a76fbb9e0f43ebde1f69  0.0s
 => [internal] load build context                                                     0.1s
 => => transferring context: 660B                                                     0.0s
```

Windo...

RAM 1.05 GB CPU 0.83% Disk: 8.88 GB used (limit 1006.85 GB) | v4.60.1

```
Terminal
 => => naming to docker.io/library/python11                                    0.0s
PS D:\docker\dockerpython> docker run -d --name p1 -p 5001:5001 python11
3e036ce702fadd460dff922e90e6924daeaa11a818a3fa250a80126a388f2083
PS D:\docker\dockerpython> docker build -t python11 .
>>
[+] Building 2.2s (11/11) FINISHED                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                           0.0s
 => => transferring dockerfile: 365B                                           0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim            1.8s
 => [auth] library/python:pull token for registry-1.docker.io                  0.0s
 => [internal] load .dockerignore                                              0.0s
RAM 1.06 GB  CPU 3.27%   Disk: 8.88 GB used (limit 1006.85 GB)                 v4.60.1
```

```
tcp, [::]:5001->5001/tcp       p1
7db0e0c17daa   mongo:latest   "docker-entrypoint.s…"   3 days ago      Up 19 minutes   0.0.0.0:27017->2701
7/tcp, [::]:27017->27017/tcp   mongodb
PS D:\docker\dockerpython> docker stop p1
p1
PS D:\docker\dockerpython>
PS D:\docker\dockerpython> docker rm p1
p1
PS D:\docker\dockerpython> docker run -d -p 5001:5001 python11
094f01b6e3ef8549923abb82d6b11da19faa5ea7d6620cf5db373b6fcb7fb4e6
PS D:\docker\dockerpython>
RAM 1.06 GB  CPU 3.09%   Disk: 8.88 GB used (limit 1006.85 GB)                 v4.60.1
```

localhost:5001

🚀 Docker Flask Server Running

Welcome to the Advanced Flask Application

- About
- Current Time
- Dynamic User
- Contact Form
- API Status

Convert the given Dockerfile to compose file

```
From httpd:latest
COPY html/  /usr/local/apahe2/htdocs/
```

1. Mkdir compose
2. Cd compose
3. Notepad docker-compose.yml
4. Mkdir html
5. Cd html
6. Notepad index.html
7. Cd ..
8. Docker compose up -d
9.